Mauro Artigiani

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Academic Positions

2024–Present **Associate Professor** (Profesor Asociado)
Universidad Nacional de Colombia, Sede Bogotá, Colombia.

2022–2024 **Graduate School Director**
School of Engineering, Science and Technology
Universidad del Rosario, Bogotá, Colombia.

2020–2024 **Principal Professor** (Profesor Principal)
School of Engineering, Science and Technology
Universidad del Rosario, Bogotá, Colombia.

2018–2019 **Postdoctoral Researcher**
Universidad de los Andes, Bogotá, Colombia.

2016–2018 **Postdoctoral Researcher**
Centro di Ricerca Matematica Ennio De Giorgi, *Scuola Normale Superiore*, Pisa, Italy.

Education

2012–2016 **Ph.D. in Mathematics**, *University of Bristol*, UK.
Thesis: *Some aspects of Diophantine approximation and ergodic theory of translation surfaces*.
Advisor: Prof. Corinna Ulcigrai.

2010–2012 **M.Sc. in Mathematics**, *University of Rome "Sapienza"*, Italy.
110/110 *cum laude*. Major: Algebra & Geometry.
Advisor: Prof. Gabriele Mondello.

2007–2010 **B.Sc. in Mathematics**, *University of Rome "Sapienza"*, Italy.
110/110 *cum laude*. Major: Pure Mathematics.
Advisor: Prof. Andrea Sambusetti.

Grants and Fellowships

2021 Teaching and Research Stay Grant (Beca de Estancia de Docencia e Investigación).

2021 Invited Professor, *University of Pisa*.

2019 Structured Quartet Research Ensemble (SQuaRE) Grant, *American Institute of Mathematics* (joint with A. Randecker, C. Sadanand, F. Valdez, G. Weitze-Schmithüsen).

2012–2016 Postgraduate Scholarship, *University of Bristol*.

Publications & Preprints

Preprints

[1] Mauro Artigiani and Angel Pardo. "Counting Cylinders on Z-covers of Genus 2 Square-tiled Surfaces". 2026. arXiv: 2610.06609 [math.DS].

[2] Mauro Artigiani, Roberto Castorrini, Davide Ravotti, and Yuriy Tumarkin. "Maharam-Pollicott-Ruelle resonances and self-similar translation flows on abelian covers". 2026. arXiv: 2607.11482 [math.DS].

[3] Mauro Artigiani, Artur Avila, Sébastien Ferenczi, Pascal Hubert, and Alexandra Skripchenko. "Typical Weak Mixing and Exceptional Spectral Properties for Interval Translation Mappings". 2026. arXiv: 2603.19401 [math.DS].

[4] Mauro Artigiani, Anja Randecker, Chandrika Sadanand, Ferrán Valdez, and Gabriela Weitze-Schmithüsen. "Realizing groups as symmetries of infinite translation surfaces". 2026. arXiv: 2311.00158 [math.GT]. To appear in Algebraic & Geometric Topology.

[5] Mauro Artigiani, Anja Randecker, Chandrika Sadanand, Ferrán Valdez, and Gabriela Weitze-Schmithüsen. "Veech groups of covers of the Chamanara surface". 2025. arXiv: 2504.00956 [math.GT].

Published Papers

[6] Mauro Artigiani, Flaminio Flamini, and Davide Ravotti. "On rigidity properties of time-changes of unipotent flows". *Ann. Sc. Norm. Super. Pisa, Cl. Sci. (5)* 27.3 (2026), pp. 1593–1648. ISSN: 0391-173X. DOI: 10.2422/2036-2145.202302_001.

[7] Mauro Artigiani, Pascal Hubert, and Alexandra Skripchenko. "Renormalization for Bruin-Troubetzkoy ITMS". *Discrete Contin. Dyn. Syst.* 47 (2026), pp. 519–547. ISSN: 1078-0947. DOI: 10.3934/dcds.2025127.

[8] M. Artigiani, C. Fougeron, P. Hubert, and A. Skripchenko. "A note on double rotations of infinite type". *Trans. Mosc. Math. Soc.* 2021 (2021), pp. 157–172. ISSN: 0077-1554. DOI: 10.1090/mosc/311.

[9] Mauro Artigiani. "Octagonal continued fraction and diagonal changes". *Rend. Ist. Mat. Univ. Trieste* 53 (2021). Id/No 9, p. 28. ISSN: 0049-4704. DOI: 10.13137/2464-8728/32167.

[10] Mauro Artigiani, Luca Marchese, and Corinna Ulcigrai. "Persistent Hall rays for Lagrange spectra at cusps of Riemann surfaces". *Ergodic Theory Dyn. Syst.* 40.8 (2020), pp. 2017–2072. ISSN: 0143-3857. DOI: 10.1017/etds.2018.143.

[11] Mauro Artigiani. "Exceptional ergodic directions in Eaton lenses". *Isr. J. Math.* 220.1 (2017), pp. 29–56. ISSN: 0021-2172. DOI: 10.1007/s11856-017-1509-8.

[12] Mauro Artigiani, Luca Marchese, and Corinna Ulcigrai. "The Lagrange spectrum of a Veech surface has a Hall ray". *Groups Geom. Dyn.* 10.4 (2016), pp. 1287–1337. ISSN: 1661-7207. DOI: 10.4171/GGD/384.

Miscellaneous

[13] Mauro Artigiani. "Cómo aprendí a dejar de preocuparme y a amar la internacionalización del currículo". In: *Conectando Mundos*. Ed. by Fray Martín Martínez Páez, Rafael Alberto Méndez-Romero, and Clara Inés García Blanco. Bogota: Editorial Universidad del Rosario, 2024, pp. 303–320. ISBN: 9789585003545. DOI: 10.12804/urosario9789585003569.

Advising

2024–2025	**Juliana Bermúdez Guzmán**, B.Sc. Thesis. *Universidad del Rosario*.
2018–2020	**Daniel Pallejá López**, M.Sc. Thesis. *Universidad de los Andes*.

Service to the Profession

Editor	Boletín de Matemáticas (2025–Present)
Peer Review	Referee for *Adv. Math., Ann. Sc. Norm. Super. Pisa, Cl. Sci., Bull. Am. Math. Soc., Comment. Math. Helv., Complex Manifolds, Ergodic Theory Dyn. Syst., Ill. J. Math., J. Mod. Dyn., Math. Commun., Stud. Math.* Reviewer for zbMATH Open and MathSciNet.
Organization	Co-organizer, Weekly Colloquium Seminar (2019), *Universidad de los Andes*. Co-organizer, "From Order to Chaos" (2018), *CRM Ennio De Giorgi*. Co-organizer, Dynamical Systems Seminar (2017–2018), *SNS Pisa*. Co-organizer, MINGLE (2015) *University of Bristol*.
Institutional	Graduate School Director (2022–2024), *Universidad del Rosario*.
Outreach & Public Engagement	Matemáticas de las redes sociales (y la vida diaria). Organized by *Ciencia café pa' sumercé*. Available on YouTube. Google, las Matemáticas, y los Sistemas Dinámicos. Available on YouTube.

Selected Teaching

2024–Present	**Universidad Nacional de Colombia** Courses: Fundamentals of Mathematics, Complex Variable, Riemann Surfaces, Topology, Groups and Rings.
2020–2024	**Universidad del Rosario** Courses: Theory of Computation, Abstract Algebra and Coding, Introduction to Logic and Set Theory, Differential and Computational Geometry, Calculus II.
2021	**University of Pisa** Graduate Course: Translation Surfaces (Ph.D. program).
2018–2020	**Universidad de los Andes** Courses: Dynamical Systems, Ergodic Theory, Calculus I & II.

Selected Invited Talks

Jul 2026	Moduli, dynamics, and integrability, *Higher School of Economics*, Moscow, Russia.
Jun 2026	Frontiers in Surface Topology: Compact, Non-Compact, and Beyond, *Casa Matemática Oaxaca*, México.
Jan 2026	IV Encuentro de Superficies en la frontera, *Universidad Nacional de Colombia*, Manizales, Colombia.
Jul 2025	Special Session: Randomness in Low-Dimensional Geometry, *Mathematical Congress of the Americas*, Miami, USA.

Jun 2025	Blooming Beasts: A Conference on the Geometry, Topology, and Dynamics of Infinite-Type Surfaces, *Casa Matemática Oaxaca*, Mexico.
Oct 2024	Dynamical Systems Seminar, *Higher School of Economics*, Moscow, Russia.
Jun 2024	Dynamical Systems Seminar, *CRM Ennio De Giorgi, SNS*, Pisa, Italy.
Jun 2023	Semi-plenary Talk, *XXIII Congreso Colombiano de Matemáticas*, UPTC, Tunja.
Dec 2022	Combinatorial and Geometric Approaches on Dynamics, *Centro de Modelamiento Matemático, Universidad de Chile*.
Jun 2022	Courant Institute Dynamical Systems Seminar, *New York University*, USA.
Jun 2021	Ergodic Theory Workshop, *University of North Carolina*, Chapel Hill, USA (virtual).
Feb 2018	Workshop around Dynamical Systems, *Udine*, Italy.
Dec 2017	Dynamical Systems Seminar, *Institut de Mathématiques de Marseille*, France.
Nov 2017	Horowitz Seminar, *Tel Aviv University*, Israel.
Aug 2017	Dynamics in Number Theory and Geometry, *Fields Institute*, Kingston, Canada.
Jun 2017	Workshop on Moduli Spaces, *SkolTech & HSE*, Moscow, Russia.
Nov 2016	Prima Giornata DinAmica, *Gran Sasso Science Institute*, L'Aquila, Italy.
Nov 2015	Simons workshop on Translation surfaces and Dynamics, *Mathematical Research and Conference Center*, Bedlewo, Poland.
Feb 2015	Ergodic Theory and Dynamical Systems seminar, *Warwick University*, UK.

LANGUAGES

Italian (Native), English (Fluent), Spanish (Fluent).